DRAFT SOLUTION

EXAM TFE 4164 Nanoelectronics 1

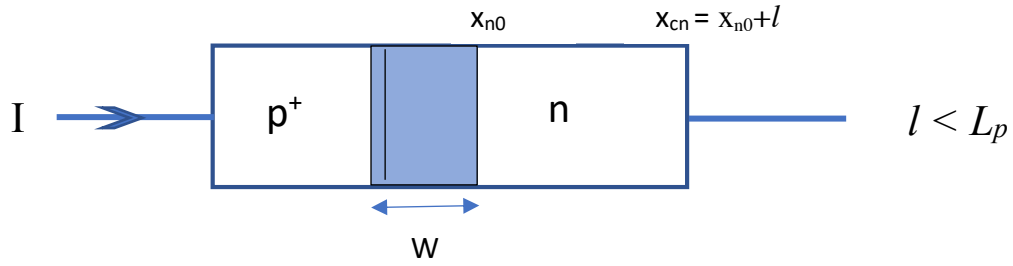
Monday 14 December 2018

Problem set I (Multiple choice)

Alternative / Problem no.	A	B	C
I-1		X	
I-2	X		
I-3			X
I-4			X
I-5		X	
I-6			X
I-7		X	
I-8		X	
I-9		X	
I-10	X		
I-11			X
I-12			X

Problem set II (p-n junction)

Q II-1



In a one-sided abrupt p^+-n junction, the injected hole current in the neutral (n-type) base region will dominate the overall diffusion current through the diode. We may thus neglect the contribution from the injected electron current into the p^+-region.

The steady state diffusion equation for the injected excess hole concentration $\delta p(x_n)$ in the neutral base region reads,

$$\frac{d^2\delta p(x_n)}{dx_n^2} = \frac{\delta p(x_n)}{D_p\tau_p} = \frac{\delta p(x_n)}{L_p^2}$$

since $\frac{\partial}{\partial t}\delta p(x_n) \equiv 0$ under steady state conditions.

This equation has the general solution,

$$\delta p(x_n) = Cexp\left(-\frac{x_n}{L_p}\right) + Dexp\left(\frac{x_n}{L_p}\right)$$

The boundary conditions are,

$$\delta p(x_n=0) = C + D = \Delta p_n$$

$$\delta p(x_n=l) = Cexp(-l/L_p) + Dexp(l/L_p) = 0$$

From these equations, we find for the integration constants C and D,

$$C = \frac{\Delta p_n exp\left(l/L_p\right)}{exp\left(\frac{l}{L_p}\right) - exp\left(-\frac{l}{L_p}\right)}$$

$$D = \frac{-\Delta p_n exp\left(-l/L_p\right)}{exp\left(\frac{l}{L_p}\right) - exp\left(-\frac{l}{L_p}\right)}$$

We thus have,

$$\delta p(x_n) = \frac{\Delta p_n \left[exp\left(\frac{l - x_n}{L_p}\right) - exp\left(-\frac{l - x_n}{L_p}\right)\right]}{exp\left(\frac{l}{L_p}\right) - exp\left(-\frac{l}{L_p}\right)}$$

The diode current is given by,

$$I = JA = -qAD_p \left.\frac{d\delta p(x_n)}{dx_n}\right|_{x_n=0}$$

$$I = -\frac{qAD_p\Delta p_n}{L_p}\left[-\frac{exp\left(\frac{l}{L_p}\right) + exp\left(-\frac{l}{L_p}\right)}{exp\left(\frac{l}{L_p}\right) - exp\left(-\frac{l}{L_p}\right)}\right]$$

$$I = \frac{qAD_p}{L_p}\left[tanh\left(\frac{l}{L_p}\right)\right]^{-1} p_n\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

To first order,

$$tanh\left(\frac{l}{L_p}\right) = \frac{l}{L_p}$$

From this we have,

$$I = \frac{qAD_p}{L_p}\cdot\frac{L_p}{l}\cdot p_n\left[exp\left(\frac{qV_a}{kT}\right) - 1\right]$$

Moreover, from the "*law of mass action*" we have, $p_n n_n = n_i^2$

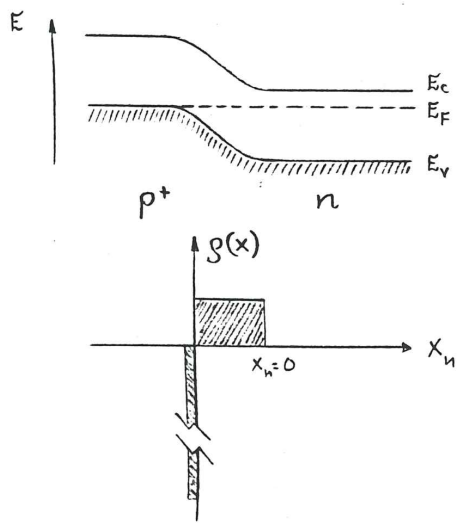
Assuming complete ionization, $n_n = N_D$

We then find,

$$I = \frac{qAD_p}{l}\cdot\frac{n_i^2}{N_D}\cdot\left[exp\left[\left(\frac{qV_a}{kT}\right) - 1\right]\right]$$

$$\boldsymbol{I = qAn_i^2\frac{D_p}{N_D l}exp\left[\left(\frac{qV_a}{kT}\right) - 1\right]} \qquad \textbf{q.e.d.}$$

Q II-2



Energy band diagram

Depletion layer charge distribution

From Q II-1 we have,

$$I = qAn_i^2 \frac{D_p}{N_D l} exp\left[\left(\frac{qV_a}{kT}\right) - 1\right]$$

where, $l = x_{cn} - x_{n0}$. In a p^+-n junction, $x_{n0} \approx W$, so that,

$$l \approx x_{cn} - W = x_{cn} - \sqrt{\frac{2\varepsilon_s(V_0 - V_a)}{qN_D}}$$

Where V_0 is the built-in equilibrium (zero bias) contact potential for the p^+-n junction. With degeneration-doped p-side, we have for V_0,

$$V_0 \approx \frac{E_g}{2q} + \frac{kT}{q} ln\left(\frac{N_D}{n_i}\right)$$

$V_0 = 0.555 + 0.026 \cdot \ln(6.67 \cdot 10^6)$

$= 0.555 + 0.409$

$= 0.964$ V

Solving the equation above for the diode current w.r.t. V_a, we find,

$$V_a = \frac{kT}{q} ln\left(1 + \frac{I}{qAn_i^2} \cdot \frac{lN_D}{D_p}\right)$$

Setting I = 1 mA and l_0 = 2 μm in this expression for V_a, we get,

$$V_a = 0.026\ ln\left(1 + \frac{10^{-3}}{1.6 \cdot 10^{-19} \cdot 10^{-4} \cdot 1.5^2 \cdot 10^{20}} \cdot \frac{2 \cdot 10^{-4} \cdot 10^{17}}{20}\right)$$

$= 0.026\ ln(1 + 2.78 \cdot 10^{11}) = 0.685$ V

l = 2 μm for V_a = 0. From the equation above relating l to the applied forward bias V_a, we have for the revised neutral base length, l', with V_a = 0.685 V,

$$l' = l + \sqrt{\frac{2\varepsilon_s V_0}{qN_D}} - \sqrt{\frac{2\varepsilon_s (V_0 - V_a)}{qN_D}} = 2 \cdot 10^{-6} + \sqrt{\frac{2 \cdot 11.8 \cdot 8.85 \cdot 10^{-14} \cdot 0.964}{1.6 \cdot 10^{-19} \cdot 10^{17}}} - \sqrt{\frac{2 \cdot 11.8 \cdot 8.85 \cdot 10^{-14}(0.954 - 0.685)}{1.6 \cdot 10^{-19} \cdot 10^{17}}}$$

$$l' = 2 \cdot 10^{-4} + 0.112 \cdot 10^{-4} - 0.059 \cdot 4 = 2.053\ \mu m$$

With this revised value l' for the neutral base length, valid for a forward bias of 0.565 V, we find,

$$V_a = \frac{kT}{q} ln\left(1 + \frac{I}{qAn_i^2} \cdot \frac{l'N_D}{D_p}\right)$$

$$\boldsymbol{V_a} = 0.026 ln\left(1 + \frac{10^{-3}}{1.6 \cdot 10^{-19} \cdot 10^{-4} \cdot 1.5^2 \cdot 10^{20}} \cdot \frac{2.053 \cdot 10^{-4} \cdot 10^{17}}{20}\right)$$

$= 0.026\ ln(2.851 \cdot 10^{11}) =$ **0.686 V**

Further iterations (recalculating l') would only have marginal impact on the value for the forward bias voltage, V_a.

Problem set III (MOSFET)

Q III-1

The energy difference $q\phi_F=E_i-E_{Fs}$ can be found from the *thermal equilibrium* hole conc. p_0 for an acceptor-doped semiconductor, assuming complete ionization (idéal MOS structure),

$$p_0 = n_i exp\left(\frac{E_i-E_F}{kT}\right) = N_A$$

Thus,

$$q\phi_F = E_i - E_F = kTln\left(\frac{N_A}{n_i}\right) = 0.0259 \cdot ln\left(\frac{4\cdot10^{15}}{1.5\cdot10^{10}}\right) = 0.0259 \cdot 12.49 = \mathbf{0.323\ eV}$$

The transistor then enters strong inversion for a surface potential, (i.e., the surface potential at which the electron conc. at the oxide interface is identical with the hole conc. in the neutral substrate).

$$\phi_s = 2\phi_F = 2 \cdot 0.323 = \mathbf{0.646\ V}$$

The depletion charge Q_d [C/cm^2] on the border of strong inversion is given by the space charge in the depletion region, $\rho = -qN_A$ [C/cm^3] over a width W corresponding to a surface potential ϕ_s at the semiconductor/oxide interface. We find this charge from Poisson's eq. (assuming the problem is solved in 1D),

$$\frac{dE(x)}{dx} = \frac{q}{\in}(-N_A)$$

which is solved for *E*(x) by integration over the depletion layer W,

$$E(x) = -\frac{qN_A}{\in}x + A(A - integration\ constant)$$

Setting x=0 at substrate end of the depletion region, the boundary condition for the electric field at this point is *E*(0)=0, (zero electric in the neutral substrate). This implies A=0. From the definition of the (built-in) electric field we have,

$$E(x) = -\frac{d\phi_s}{dx}$$

$$\int_0^W E(x')dx' = \int_0^W -\frac{qN_A}{\in}x'dx' = \int_{\phi_s}^0 d\phi_s' = -\phi_s$$

$$\phi_s = \frac{1}{2}\cdot\frac{qN_A}{\in}W^2$$

$$W = \left(\frac{2\in\phi_s}{qN_A}\right)^{\frac{1}{2}} = \left(\frac{2\epsilon_r^{sc}\epsilon_0\phi_s}{qN_A}\right)^{\frac{1}{2}}$$

We then have,

$$Q_d = -qN_AW = -(2\epsilon_r^{sc}\epsilon_0\phi_s qN_A)^{\frac{1}{2}}$$

$$= -(2 \cdot 11.8 \cdot 8.85 \cdot 10^{-14} \cdot 0.646 \cdot 1.6 \cdot 10^{-19} \cdot 4 \cdot 10^{15})^{\frac{1}{2}}$$

$$= -2.939 \cdot 10^{-8} = \mathbf{-29.4\ nC/cm^2}$$

The threshold voltage V_T for the transistor is the voltage applied to the gate contact relative to the substrate in order to obtain the surface potential ϕ_s at the semiconductor/oxide interface, corresponding to a charge Q_d on the semiconductor side of the MOS capacitor. With a capacitance C_i [C/cm^2] for the gate oxide,

$$(V_T - \phi_s)C_i = -Q_d$$

where,

$$C_i = \varepsilon_r^{ox}\varepsilon_0/d = 3.8 \cdot 8.85\times10^{-14}/10^{-5}\ \text{C/cm}^2$$

Thus,

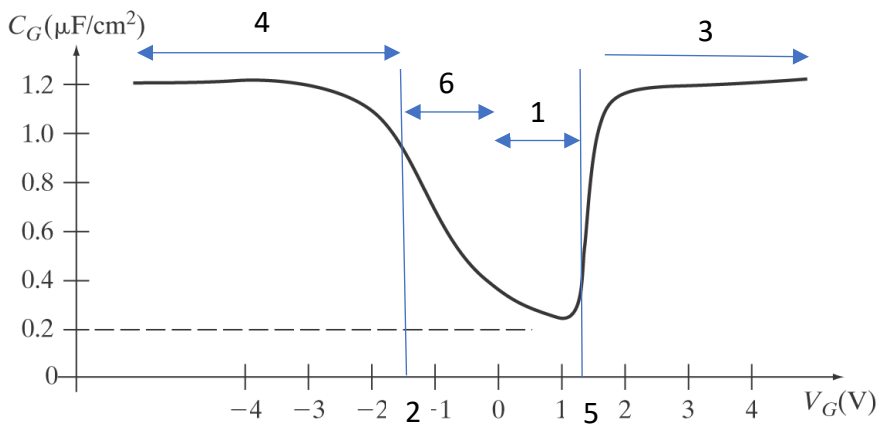
$$V_T = \phi_s - \frac{Q_d \cdot d}{\varepsilon_r^{ox}\varepsilon_0} = 0.646 + \frac{29.4\cdot10^{-9}\cdot10^{-5}}{3.8\cdot8.85\cdot10^{-14}} = \mathbf{1.520\ V}$$

The imposed gate voltage corresponding to flat bands in the semiconductor may differ from V_G=0 because of,

i) different work function for semiconductor and gate oxide

ii) fixed oxide and (semiconductor/oxide) interface charges

Q III-2



The oxide capacitance, C_{ox}, is the capacitance in strong accumulation, i.e., for negative applied gate voltages, V_G. Thus,

C_{ox} = **1.2 μF/cm²**

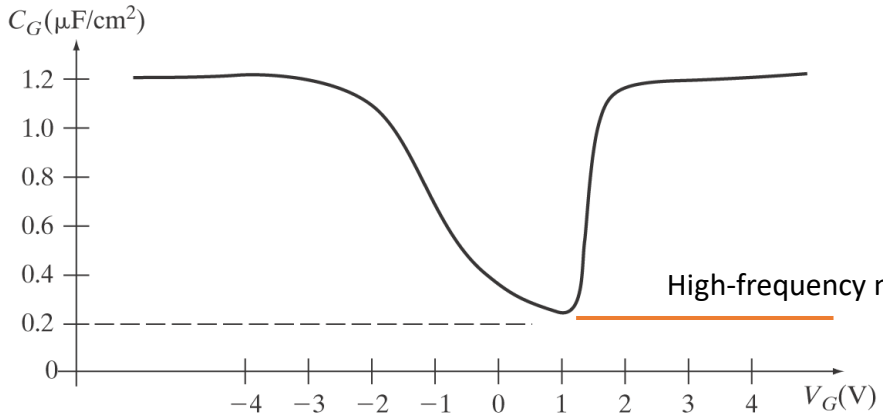
The minimum depletion capacitance, C_d^{min}, is the depletion capacitance corresponding to the maximum depletion width, W_m (the depletion width at the onset of strong inversion). The minimum MOS capacitance, C_{min} is the series combination of C_{ox} and C_d^{min},

C_{min} = 0.2 μF/cm²

$$C_{min} = \frac{C_{ox} \cdot C_d^{min}}{C_{ox} + C_d^{min}} = \frac{1.2 C_d^{min}}{1.2 + C_d^{min}} = 0.2$$

$$C_d^{min} = \frac{0.2}{1.2}\left(1.2 + C_d^{min}\right) = 0.167\left(1.2 + C_d^{min}\right) = 0.2 + 0.167 \cdot C_d^{min}$$

$$C_d^{min} = \frac{0.2}{1 - 0.167} = \frac{0.2}{0.833} = \mathbf{0.24\ \mu F/cm^2}$$



Q III-3

This MOSFET would be a *normally ON/depletion mode* device, since we would have to apply a negative gate voltage, V_G, to turn the transistor OFF (negative threshold voltage, V_T).

This MOSFET would be a long channel device, since the I_D–V_D characteristics are flat (independent of V_D) beyond "pinch-off". No *Early* effect"!
This conclusion may also be reached from the separation of the saturation regime I_D–V_D characteristics for a given increment in VG, which should increase as $(V_G–V_T)^2$ for a long-channel MOSFET and as $(V_G–V_T)$ for a short-channel device.

It is seen from the I_D–V_G characteristic that the *subthreshold slope/swing* is,

$S = dV_G/d(\log I_D)$ = **25 mV/decade**

The subthreshold slope can be written,

$$S = \frac{kT}{q} ln10 \left[1 + \frac{C_d + C_{it}}{C_{ox}}\right] = 0.0259 \cdot 2.3 \cdot \left[1 + \frac{C_d + C_{it}}{C_{ox}}\right]$$

$$= 59.6 \cdot \left[1 + \frac{C_d + C_{it}}{C_{ox}}\right] \text{ mV/decade}$$

Where C_d is the depletion capacitance, C_{it} is the capacitance from interface trapped charges, and C_{ox} in the gate oxide capacitance. These are all positive numbers in the formula above. Thus, S has a minimum value of approximately ***60 mv/decade***, and so apparent value for S (25 mV/decade) is unphysical.